MUO - Muon Lifetime

Physics 111B: Advanced Experimentation Laboratory

University of California, Berkeley

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1 Description of the Muon Lifetime Experiment (MUO)

Revised 2026-8-20 by Auden Young.

1. **Note that there is NO eating or drinking in the 111-Lab anywhere, except in rooms 282 & 286 LeConte on the bench with the BLUE stripe around it.** Thank You the Staff.

The muon is an unstable particle, with a lifetime on the order of microseconds, which decays by means of the weak interaction into an electron and two neutrinos. If we start out with N_o muons, then the number of muons present at a time t later is $N(t) = N_0 e^{-\frac{t}{T}}$ where T is the mean life.

In this experiment, cosmic-ray muons that enter a large tank of liquid scintillator trigger the emission of a pulse of photons. These photons are detected by a photomultiplier tube. If the muon stops inside the tank,

another pulse of light is produced when the muon decays. We can use the distribution of time differences to measure the muon lifetime. A histogram of these time differences for many decaying muons resembles an exponential distribution from which the mean lifetime can be determined.

You will get acquainted with a number of electronic devices that are quite common in laboratories. Data analysis using a computer and program writing are required. You will spend time in setting up the equipment. Once that's done and data-taking is started, you don't need to be in the lab, except for checking out how everything is working.

1. Pre-requisites: None
2. Days Allotted for the Experiment: 6

All pages in this lab. Note To print Full Lab Write-up click on each link below and print separately

1. **Exponential Probability Distributions**
2. **Calibration Setup Procedure**
3. **Muon Analysis in ROOT**

This lab will be graded 20% on theory, 20% on technique, and 60% on analysis.

2 Muon Lifetime Experiment Photos

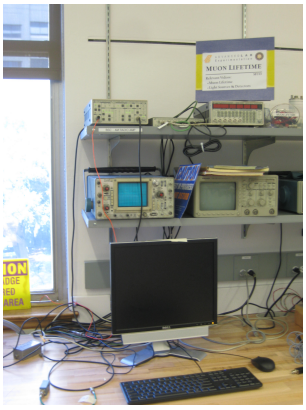


Figure 1: Front view of the lab station for the muon lifetime experiment. See larger image **here**



Figure 2: High voltage power supplies for top & bottom PMTs with the black base of the top PMT shown. See larger image **here**



Figure 3: Close up of the black base of the top photomultiplier tube found in room 275. See larger image **here**



Figure 4: Vertical shaft found in room 275, where the large liquid scintillation tank sits. See larger image **here**

3 Before the 1st Day of Lab

Complete the MUO Pre-lab found in the Signature Sheet for this lab. Print the signature sheet, discuss the experiment and pre-lab questions and answers with any faculty member or GSI, and receive their signature. In the course of the lab there will be examination points where you must STOP and get a GSI or professor to verify your understanding and/or verify proper experimental setup. You cannot skip these checkpoints, and must receive signatures demonstrating that you've consulted the staff. Some experiments may have mid lab questions that must be completed by specific days of the experiment. The completed Signature Sheet MUST be submitted as the first page of your lab report. Quick links to the checkpoint questions are found here: 1 2 3 4 5

1. ***Note: In order to view the private Youtube videos hosted by the university, you must be signed into your berkeley.edu Google account.***
   View the **Muon Lifetime Video**.
2. Last day of the experiment please fill out the **Experiment Evaluation**

Suggested Reading:

1. B. Rossi, "**Interpretation of Cosmic-Ray Phenomena**", Rev. Mod. Phys. **20**, 537 (1948).*Gives the cosmic ray flux at sea level. Figure 6 on p. 545 is important for your calculations.*
2. D. Griffiths, ***Introduction to Elementary Particles***, 1987.*Read the introduction (pp. 1-10). The result of a tedious calculation of the muon lifetime is given in pp. 307-309.*
3. R.D. Evans, ***The Atomic Nucleus***, McGraw-Hill (1955).*Read about radioactive decay in Chapter 15.*
4. C.S. Sutton et al., "**Undergraduate Cosmic Ray Muon Decay Experiments with Computer Interfacing**", Computers in Physics, Nov/Dec, 76 (1987), #QC52.C658.
5. O.C. Allkofer, ***Introduction to Cosmic Radiation***, Verlag Karl Thiemig, Munchen (1975). Read Sections 4.1, 9.1-9.6, 9.12, and 10.1-10.4.
6. "**RCA-4522 Photomultiplier Tube Specifications**", RCA Electronic Components, June, 1 (1968).
7. F. Riggi, *Messengers from the Cosmos: An Introduction to the Physics of Cosmic Rays in Its Historical Evolution* - available online via the Berkeley library; very helpful for the additional investigations.

More References and Manuals

You should keep a laboratory notebook. The notebook should contain a detailed record of everything that was done and how/why it was done, as well as all of the data and analysis, also with plenty of how/why entries. This will aid you when you write your report.

4 Objectives

- Measure muon lifetime and observe the effects of special relativity
- Measure local muon flux
- Learn how to analyze a convenient source of genuinely random numbers
- Understand sources of error in detectors and processes involved in detector calibration
- Gain familiarity with computational tools of particle physics like ROOT
- Simulate muons and measure their simulated lifetime using Monte Carlo methods
- Learn data analysis tools, fitting histograms, and other methods of event processing
- Design an investigation of your choosing and develop a plan to carry it out

5 Introduction

The goal for this lab is to measure the lifetime of the muon. This is a relatively simple experiment, but one that can achieve surprisingly good precision if you are careful. Calibrating the electronics properly is important and is worth doing carefully. Most of the adjustments are made in the analysis using software. Don't rush through the adjustments, but ensure you understand what you are doing and how your changes affect the data. Get the settings right before spending all day and night recording data; otherwise you only find out later that the data are useless. Once you have the data and a nice plot on the screen, you still need to work hard to make a proper analytical fit and to extract a correct value for the lifetime.

Muons are particles that have a mass over 200 times that of the electron. Muon decay is of great importance in the study of weak interactions, one of the fundamental forces in nature. The value of the muon lifetime is one of the most precisely known constants, and is currently measured with a precision of a few parts per million. In this experiment, low-energy muons are primarily created from decays of particles produced in the interactions of high-energy cosmic-rays with the earth's atmosphere. Eighty percent of the cosmic rays at sea level are muons, positively or negatively charged. They have a mean lifetime of a few microseconds.

They decay by the following processes:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu \tag{1}$$

and

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu \tag{2}$$

By the CPT theorem, the lifetimes of the negative and positive muons in *vacuum* (i.e. due to reactions (1) and (2)) are equal. However, in matter, negative muons, when brought to rest in the close neighborhood of some nucleus (Z,A) a muonic atom is formed with the muon orbiting in the K-shell. The negative muon in the muonic atom can either decay as shown in reaction (2), or it can disappear through the competing reaction:

$$\mu^- + p \rightarrow \nu_\mu + n \tag{3}$$

where p and n are the proton and neutron respectively. The rate of the muon-capture process (3) depends strongly on the charge Z of the nucleus which captures the muon. In muonic carbon atoms, the rate of the muon capture (3) is about 10% the decay rate in (2).

Most of the muons that are incident on the detector pass straight through it. Only the lowest energy particles come to rest within the scintillating liquid itself.

6 Apparatus

6.1 Overview

The large liquid scintillation counter is located in a shaft which can be accessed from room 275 Physics North. It is filled with 125 gallons, 500 kg (1/2 cubic meter) of mineral oil in which a scintillating substance is dissolved. The passage of fast charged particles through the liquid gives rise to short pulses of light a few nanoseconds in duration. These flashes of light called scintillations are converted into electrical pulses by photomultiplier tubes located at the top and bottom of the detector (Figure 5). Each photomultiplier views half of the tank, which is divided by an opaque baffle in the horizontal mid plane.

A photomultiplier tube (PMT) is a detector that converts single photons into large pulses of electric current by successive multiplying stages (Figure 6).

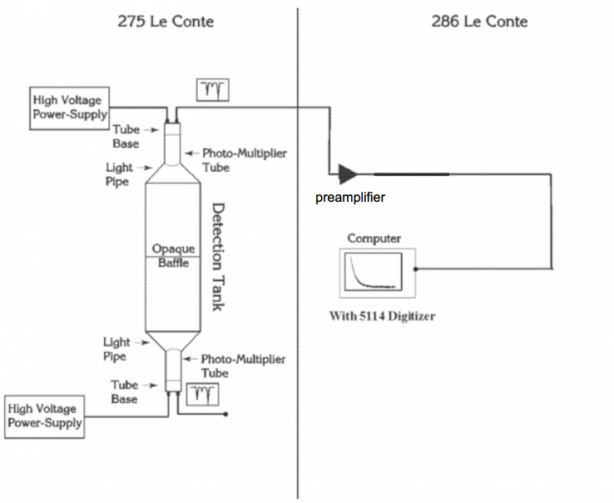


Figure 5: Block diagram of the muon lifetime experiment; the computer now uses the ADS as a digitizer instead of the NI 5114 digitizer card.

In the shaft of room 275 Physics Building (Get a GSI to let you into the room) **are located:**

- a high voltage supply and a high voltage divider for the photomultiplier tubes. The high voltage of the photomultiplier tube normally is left on and is NOT adjusted by the student. See the staff in the 111-LAB. These photomultiplier tubes are irreplaceable. Do not mess with them.
- a voltage meter to read the individual high voltages. ***Under no circumstances should the high voltage exceed 2200 V!***
- one large liquid scintillator with 500 kg of doped mineral oil
- cables leading to 286 Physics North where the rest of the apparatus is located.

At the 286 Physics North Muon Lifetime Station are located:

- SR445A 4-channel x5 amplifier, for amplifying the signals from the top PMT (white cable) and bottom PMT (black cable). The flashing red lights should indicate that they are receiving a signal.
- Tektronix TDS 360; Digital Storage Scope
- Tektronix 465 Oscilloscope (Analog)
- Miscellaneous equipment for calibration of the system and 50 ohm terminators
- Stanford Research Systems Model DG645 Digital Delay Generator for MUO-1 and DG535 Four Channel Digital Delay Generator for MUO-2
- Digilent Analog Digital Studio (ADS)

The ADS works together with a Python program that does two things: (1) it is an oscilloscope, (2) it is a differential detector logic system.

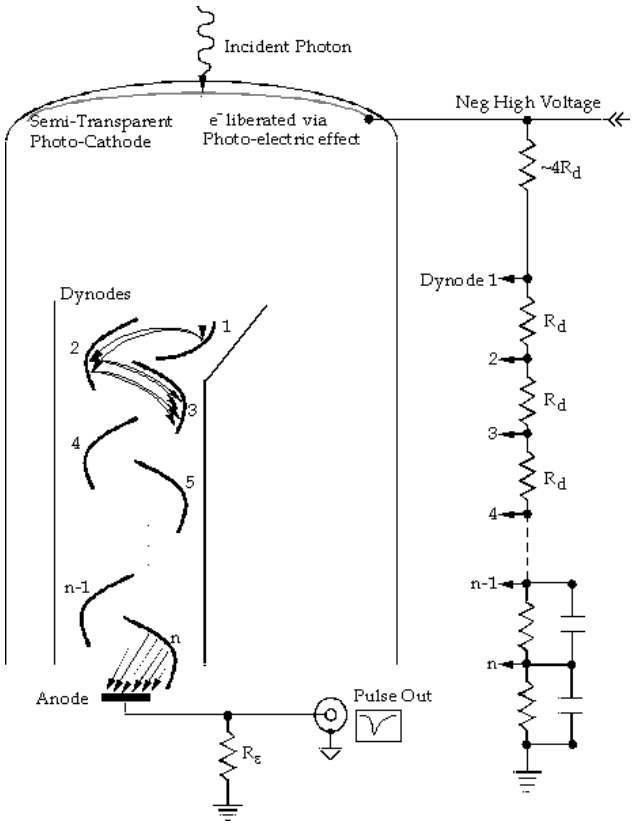


Figure 6: Schematic of a photomultiplier tube

6.2 How the Experiment Works

Muons are derived from the process of cosmic ray particles striking the nuclei of atoms and molecules in the atmosphere. A muon that passes through the liquid scintillator tank generates scintillation. A photomultiplier tube picks up the light pulse and sends an electrical signal through a cable to our experiment. A muon that enters, stops, and subsequently decays in the liquid scintillator gives rise to two pulses separated by a few microseconds, one when it enters the liquid, and a second one when it decays. The pulses can be viewed with an oscilloscope or digitized and stored on the computer. You are going to measure the time difference between these correlated pulses, as well as the amplitude and width of each pulse. After many such observations, you will obtain a distribution of the number of counts vs. time difference. Ideally, when these data are plotted, they will form an exponential decay curve, whose decay constant is the lifetime of the the muon.

When the two pulses come down the cable, they are routed to an amplifier. Then they go to the ADS. You configure triggering and timing properties of the digitizer in the Python program. The Python program **Muon Detector Program** coupled to the ADS detects pulses from two photomultiplier tubes, measures their arrival times, amplitudes, and widths, and, for a subset of pulses that satisfy a preset criteria, writes the data to disk. You will read the data offline, filter it to reject noise, plot the time difference between two pulses in the same photomultiplier tube, and use the distribution to measure the muon lifetime.

6.3 Explanation of the Python Program

- Note: High voltage on the PMTs are as follows: Top tube =1900 Volts Negative and bottom tube = 2300 Volts Negative with the SR445A 4 each amplifier, setup top PMT 1 -amp, bottom PMT 2 amps.

This experiment uses a Python program called **Muon Detection Program** with two different tabs: **View Signals** and **View Histogram**. The purpose of **View Signals** is for you to choose the best input parameters by inspecting the signals and seeing how changes in the input parameters affect the signals and counting rates. (This reads the input signals like an oscilloscope). The optimal input parameters should then be used in the program **View Histogram**, whose purpose is to save pulses to data files, from which you calculate the muon lifetime.

We use the Digilent ADS to act as a high-speed digitizer. The top PMT is connected to channel 0 of the ADS scope, and the bottom PMT to channel 1. When you start an acquisition, the scope is automatically configured based on parameters you input (like sample frequency and input range) as well as some hidden parameters like buffer size (you should read the ADS documentation to understand what these do behind the scenes). The default range setting of 1 V is about right for both channels, but you should feel free to play with these values - it defines the range of values you can see, but also the resolution with which you can see them. The bandwidth of the ADS scope is 30+ MHz. You should read about the specifications of the ADS in the manual above.

A detailed description of the two tabs and their settings is given below.

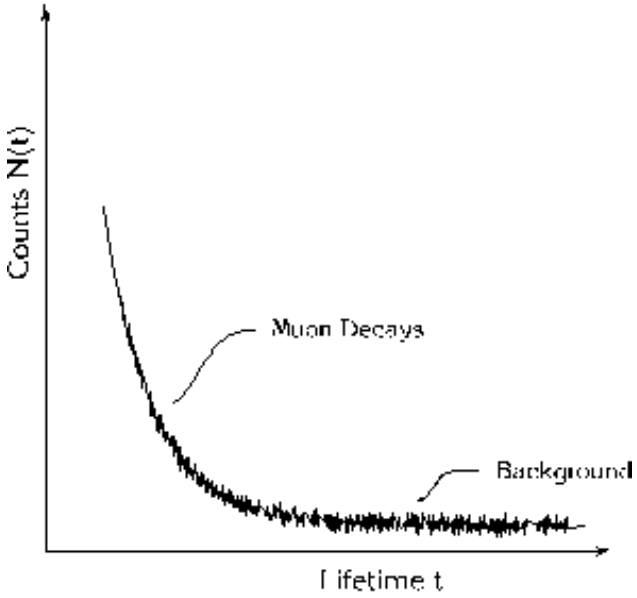


Figure 7: Distribution of the temporal separation between two signals. Note that the X axis sometimes is called the channel number for the digitizer. The Y axis is also called the amplitude counts.

6.3.1 View signals

The View Signals tab of the Muon Detection Program (see Figure 8) has several main settings, many of which carry over to the Live Histogram tab. There are three main categories of setting: channel-specific (both for channel 0 and channel 1), signal processing, and acquisition. Working from the bottom up on the side bar, the acquisition settings/controls are:

- **Sample rate (Hz)** - leave at 100 MHz - the sampling rate of the ADS scope.
- **HW trigger channel (0 or 1)** - select which channel of the ADS to trigger on.

- **Connect/disconnect** - Connect and disconnect to the ADS. When moving between tabs, you'll need to disconnect on one tab and then reconnect on the other.
- **Start/stop/clear** - Start, stop, and reset a measurement.

This also has a small status panel indicating if there is an active measurement, any errors in connecting, etc. The signal processing settings are:

- **Low pass cutoff (Hz)** - the program passes a received signal through a low pass filter in order to reduce noise. This sets the high-frequency cutoff for that filter.
- **Window start after trig (μs)** - Once the trigger is activated, we wish to ignore any pulses that occur shortly after, since they may simply be noise or part of the first pulse, as opposed to a muon decay event. This control allows the user to define how long this ignore period should last. This control should be large enough to ignore this noise, but not so large as to ignore a significant portion of the actual muon decay events.
- **Window stop after trig (μs)** - determines the length of the record (counting from the trigger), i.e. the maximum difference between the trigger and the second pulse. The triggering event is set to be at 1 μs of the way through each record, so that the digitizer can see pulses just before the trigger and so that some of the extra pretrigger data the ADS automatically collects is discarded and doesn't affect processing speed.[1] After this time has passed, the program analyzes the collected data for pulses, and then returns to listening for the next trigger event. This control should be just large enough to encompass a large majority of decay events.[2]
- **Holdoff (μs)** - The amount of time after a pulse/trigger to wait before looking to see if there is another trigger. If this is zero, the same pulse may count as multiple trigger events.
- **Max traces to display** - how many traces to view at a given time.

The channel specific settings have a panel for both channel 0 and channel 1, with a checkbox to enable channel 1. Think carefully about what you expect to be the same or different across the channels! Also recall that the hardware itself can only trigger on one channel, so only one of the 'first triggers' is actually a *hardware* trigger and the other is a *software* trigger (what does this mean, and how does it impact the experiment?). The channel specific settings are:

- **First trigger level (V)** - the signal amplitude required for the trigger to be activated (looking for the first pulse).
- **Second trigger level (V)** - upon passing the first trigger level, the record is searched to see if a second pulse within the window defined in the signal processing settings exists which has an amplitude greater than this threshold (looking for the second decay pulse).
- **Expected pulses (1 or 2)** - Once the program has identified all of the peaks existing on the most recently received signal, it must determine whether or not these pulses represent a muon decay event. The event is registered as a true muon decay event only if the number of pulses on Channel 0 and on Channel 1 match exactly with the specified **expected pulses**. There are three suggested use cases: [CH0 = 2, CH1 = 0]: This signifies a muon which entered the chamber connected to channel 0, and then decayed in that same chamber at a time at least as long as defined by **Pulse 2 delay**, but no longer than **Range**. Since there are only two pulses on this channel and no pulses on the other channel, we are convinced that another muon did not interfere with the measurement. For this setting, the user should set the trigger on Channel 0. [CH0 = 0, CH1 = 2]: Same as above, only for the chamber connected to channel 1. For this case, the user should set the trigger on Channel 1. [CH0 = 1, CH1 = 1]: This represents a muon event in which a muon is first detected in one chamber, but then its decay is detected in another chamber, and no other pulses are recorded. It is suggested that for this case,

[1] There are some downsides to this - there were a lot of choices made in the code about how to minimize deadtime and how to quickly process incoming records. Something worth thinking about and discussing with staff if you have questions!

[2] If it is longer, this increases deadtime.

the HW trigger channel is set to the channel which is connected to the chamber which is on top, since more likely than not, we expect the muon to come from the above the surface of the Earth, and not below.

- **Voltage range (V p-p)** - This sets the voltage range of the ADS, and in so doing, effectively sets its resolution. See the ADS manual for details. This value should be just large enough to encompass the amplitude of any received pulse.
- **Attenuation** - In order for the trigger to operate successfully (rising edge is assumed) a positive signal with respect to ground is expected on both channels. However, the incoming signal from the experiment may be negative, which would crash the program. Set the attenuation to -1.0 to invert the channel's signal, or to 1.0 to not invert it. Use the oscilloscope to check the signal polarity if you are not sure.

There is also a status panel that indicates how many events have been observed (i.e., records that passed the first trigger level) and how many counts/'passing triggers' there are (i.e., pulses that passed both the first and second trigger level).



Figure 8: Muon Detection Program Signal Viewer tab

Both programs initiate a record acquisition by the digitizer and, once the voltage exceeds the trigger threshold, retrieve voltage waveform data for both channels, incrementing the **events read (first trig)** count in the status panel by one. The programs remove the voltage offset by averaging the voltages of the first 200 samples of the record, all of which should occur before the triggering event, and subtracting that voltage from all the voltage waveform data. The programs then search for up to two pulses below (for -1.0 attenuation) or above (for 1.0 attenuation) the **Channel 0(1) first trigger,** which you set. This threshold is relative to the offset, not ground. The first pulse will correspond to the Muon entering the scintillating material within

the tank and the second to its decay. The time difference between the first and second pulses has to be less than the **window stop after trig**, which you set, and greater than the **window start after trig,** which you set. The range is simply a measure of how long the program will wait between pulses before moving on to the next signal. The pulse 2 delay is there in order to ignore any noise generated after the first signal that may pass the trigger level. Typically 1 microsecond is enough breathing time. You can specify how many pulses in each channel is required for the data to be written to disk; in our case we are looking for two events (i.e. the incident and the decay). The **passing triggers** is incremented if exactly that number is found in each channel. Whether or not the output condition is satisfied, the process repeats with the next data record.

It is important to note that in the program **View Signals**, after clicking **Start Acquisition**, the program continues to acquire records from the amplifier and searches them for the specified pulses until you click **Stop Acquisition**. This program graphs the **Raw signals** acquired by the digitizer, the **Signal with offset removed** and the **Filtered signal** for each enabled channel.

6.3.2 Live Histogram

Once you have determined the optimal settings for the program controls using the **Signal Viewer** tab, the next portion of the lab deals with the**Live Histogram** tab (see Figure 9) which measures the pulse height and width in addition to time.



Figure 9: Muon Detection Program Live Histogram tab

There are a few additional settings on top of the ones for the **View Signal** tab, described below:

- **Histogram bins** - Set the number of bins to use in the histogram. This is for viewing convenience as

you are performing data collection; you should be exporting the csv of data and performing your own analysis/constructing your own histogram for your report.

- **Export csv** - At any point you can export a csv of all the data collected to that point. However, it's not strictly necessary if you set a peak-data log path (below) before beginning your run (strongly recommended).
- **Set peak-data log path** - This allows you to set a file name to save data to at the beginning. The program automatically logs data at periodic intervals so that even if the computer crashes, your data up to that point is not lost.
- **Save all charged particles** - For one of the investigation options - saves data for all records that pass the first trigger, whether or not they pass the second trigger (indicates whether or not the second trigger is passed for each record).
- **Save raw traces to csv** - For one of the investigation options - saves the raw trace/record data to a separate csv (the path you set above with '_raw_trace' appended), with the regular record statistics (pulse height, FWHM, time, etc) goint to the path you set above. Do *not* select this for long runs without discussing with a staff member - you will end up with an entirely too large file and may crash the computer.
- Rate (Hz) - additional status indicator indicating the average count rate.

When you click start, the program begins searching for pulses according to the conditions you set, just like in the **Signal Viewer** tab. When an output condition is satisfied (i.e. exactly the specified number of pulses is found in each channel, up to the maximum of 2), the program writes out a record with the following values: timestamp, whether or not it passed both triggers (relevant for the 'save all charged particles' option), time, amplitude, and width of four pulses: 1st pulse in Channel 0, 2nd pulse in Channel 0, 1st pulse in Channel 1, 2nd pulse in Channel 1, time between pulses in Channel 0, time between pulses in Channel 1, and time between first pulse across channels. If a given pulse is not found in the record (or some value is not relevant, e.g. if channel 1 is not being used), the time, amplitude, and width are assigned zero values. **IMPORTANT FOR DATA ANALYSIS:** It is the values saved in this .csv file that you will be working with in the analysis of this lab. This is the file that you should be uploading to Python, MatLab, or ROOT as it will have all the information about the signals received from the PMT.

As the program runs, it updates a histogram of decay events, where the x-axis specifies the time between the two registered pulses. If everything is working correctly, this should mimic the muon lifetime decay curve.

The program also measures the amplitude and the full-width-half-max of each peak from the signal out of the filter, and plots these on histograms as well (which you should also regenerate yourself). The filter removes a lot of the noise, so the amplitude and width information are more accurately measured with a filter. Timing information is more accurately measured before the signal is filtered.

Finally, the program also measures count rate over time, displaying both an average value in the status panel and a histogram updated over the course of the run, which is used in some of the options for the final investigation. You should also look in general for anomalies in this count rate and whether that might affect your data.

7 Procedure

7.1 Inspecting the apparatus

Inspect the apparatus in Room 275: (Get GSI to let you into the room.)

- Check the voltage setting of the PMTs but do not change (the knobs on the power supply are accurate, not the power supply's meter).

- Notice the color of the co-axial cables on the output of the high-voltage amplifier (connected to the PMT)

Checkpoint **Apparatus in the other room:**↑ **Make sure you have seen the apparatus in the other room.**

7.2 Inspecting the Signal

Now return to the MUO station in the 111B Lab Room to view the signals on the oscilloscope. The purpose of this is to get a visual estimate of the pulse amplitudes and time separation between the initial interaction peak and the decay peak. You will use these estimates in the Python program.

- The cables that originate in Room 275 should be connected to the amplifier. Check the gain setting of the amplifier. The amp outputs a 5x gain.
- Using a fast digital storage oscilloscope (Tektronix MSO-2024), observe the output of the amplifiers. The cable must be terminated at the oscilloscope with a 50Ω (what happens if it is not? Try it).
- Find appropriate trigger and time-scale settings for detecting muons. You do not want to trigger on too low of a voltage, the erratic oscillations are simply noise and you are looking for very narrow sharp pulses following the initial large pulse.
- You should see pulses at ∼100-1000 Hz up to ∼1.0 volt in amplitude, followed occasionally (∼1 Hz rate) by smaller delayed pulses. Many of these are muon decays, others are the result of various sources of noise. This is where the evolutionary capabilities of your eyes and brain working together will come in handy!
- Note the polarity, typical magnitude, and width of the pulses
- Once you have all this information written down it will make optimizing the Python program a lot easier.

Checkpoint **Oscilloscope (MSO-2024):**↑ **Explain why the cable is terminated at the oscilloscope. What polarity are the signals, positive or negative? What are the pulse widths and amplitudes? The scope will trigger depending upon the polarity of the signal, it is important to understand what polarity the signals are before moving on to the Python program as it will need to be specified when the data acquisition begins. Describe how a typical muon decay process looks like in the scope and how the first and second trigger levels might affect the measurement in the program. Show a GSI your signals on the oscilloscope and discuss what values you think are appropriate for the Python program.**

7.3 Determine optimal settings using the Python program tab View Signals

Program operation and description

1. Now use the **View Signals** tab of the **Muon Detection Program** Python program on the desktop.
   - If the pulses coming into the ADS card are negative, **Attenuation** should be set to -1.0.
   - Estimate the average height of the primary and delayed pulses.
2. Inspect the amplified pulses with the **View Signals** tab. Vary the frequency cutoff and observe how the signal and noise change. Pick the frequency cutoff that optimizes signal/noise ratio. Set the thresholds so that the counting rates (both input and output) make sense. Both should be safely above the noise level.
   - For the standard data taking (i.e. when you will be collecting data for measuring the muon lifetime), you will require 2 pulses in the top PMT (Channel 0) and 0 pulses in the bottom PMT (Channel 1) (or you can keep channel 1 off). What does this setting mean?

3. **If you are in MUO-1**: You will use the pulse generator (SRS DG645) to simulate a sequence of two pulses of approximately the size and width as the output of the detector in Room 275 later for calibration. Thus, you should get familiar with using the pulse generator. The first thing you should take note of is the front panel of the SRS DG645 for producing the pulses. The "edge" section is located on the left side of the display. The "edge" refers to the 5 different output channels, each with a 50 ohm impedance, located in the bottom of the front panel under the display. You will notice that the outputs have two letters associated with them, each located on a different edge of the signal. The first edge refers to the offset above ground and the second edge to the step size, this will effectively determine the peak-to-peak voltage of your signal above ground (an example would be A and B, where A is the offset and B is the step size). Under the display there is a section labeled "display" where you can navigate the different edges and move the cursor on the display (indicated by the character that is flashing). The two signals should be connected to ports AB and CD. Adjust the voltage for the expected peak heights as discovered in Section 7.1 ("inspecting the apparatus and the signal"). Next you will want to set the delay of the second pulse relative to the first. This can be accomplished by pushing the "delay" button located in the display section. It will prompt you to input a time delay in seconds, notice the white lines and labels located right above the display, these provide you with the units of input for your delay. Simply move the cursor over to the digit you wish to change and then adjust using the up and down arrows located directly under the 'MODIFY" section. However, make sure that when you set the delay you are selecting the correct edge to delay and that the green LED light is on for the output programmed to be the second pulse (for example if you want CD to have a delay you would make sure the LED green light is located on C and adjust the delay time). The width of the pulse can also be determined by the delay function, however now the delay is not between different outputs, but rather it is between different edges of the same output. Make sure that the LED green light is located on the second edge of the output signal and adjust the delay for the desired width of the signal. Now you should be well equipped to navigate your way through the pulse generator. Use the oscilloscope to examine the generated pulses to make sure they are consistent with your settings. However if you need any more information call over a GSI or professor and also feel free to examine the manual found here: **SRS DG645 Manual**.

4. **If you are on MUO-2** You will use the pulse generator (SRS DG535) to simulate a sequence of two pulses of approximately the size and width as the output of the detector in Room 275 later for calibration. Thus, you should get familiar with using the pulse generator. Find the Menu section on the front panel of the pulse generator. Press the Trigger button. You will see a menu that looks something like this: **Int Ext Ss Bur Line**. There will be an underscore, likely under the I, that represents the location of the cursor. This can be moved with the left and right arrows on the front keypad. If the cursor is not under Int, move it so that it is. When on the Int setting, the pulse generator will send a step to the trigger output (T0) at a rate determined by an Internal clock. Press Trigger again. You will see a menu that looks something like this: **Rate = 6.000**. This is the rate at which trigger pulses are sent on the T output in pulses per second. Now the cursor indicates which of the displayed digits you are editing. Use the right and left arrows to move the cursor, and the up and down arrows to change the digit values. Press the Delay button. You will see a menu that looks something like this: **A = T + 0.008 000 000 000**. This means that 8 ms after each trigger pulse, the pulse generator will send a step to the A output. Press Delay again. You will see a menu that looks something like this: **B = A + 0.000 055 000 000**. This means that 55 microseconds after the step is sent to A, a step will be sent to B. The 'A +' means that the B output is delayed relative to A. You could change this so that B is delayed relative to T (or C or D) by editing the A just like you would edit the numbers. Continuing to press Delay allows you to set the delays for channels C and D. Note that C should be timed relative to A, and that D should be timed relative to C. In addition to the step outputs A,B,C and D, there are four pulse outputs. The output labeled with a pulse between A and B puts out a high voltage only for the period between when a step is sent to A and when a step is sent to B. I.e., with the settings above, it turns on at time T+8ms and turns off 55 microseconds later. The channel labeled with an inverse pulse between A and B is the inverse of the one with the non-inverted pulse. The CD pulse outputs are programmed just like the AB outputs. The Output menu allows you to specify the high and low voltage levels for each of the steps and pulses, as well as the expected load on

the outputs. Please refer to the SRS DG535 manual for any further questions.

Checkpoint **Trigger Rate:**↑ **With the program & apparatus counting muons, estimate the *Events read* (first trigger) rate and the *Passing triggers* (first and second trigger) rate that are displayed on the View Signals status panel. Do these rates agree with your calculations for the muon incidence and stopping rates (from the pre-lab)? If the rates do not make sense, you might try adjusting the program thresholds (what does this do?). How might changing the threshold settings affect the *accuracy* of your data? Discuss these questions and your reasoning behind their answers with a GSI.**

7.4 Data Collection

1. Set up the **Live Histogram** tab of the Muon Detection Program & apparatus. Transfer your channel settings from the **View Signals** tab to the **Live Histogram** tab. Make sure to disconnect the ADS in the View Signals tab and connect it in the Live Histogram tab.

2. For now, make sure you have de-selected 'Save all charged particles' and 'Save raw traces to csv'. **You do not need these yet!**

3. You can live update the number of bins for the histogram as the program is running, though if you do this relatively late in the process, the program may be quite laggy.

4. Set the file to save data to.

   - Before clicking 'Start', you should set the file path you want to save data to. It will save data periodically throughout the program running in case the computer crashes. You can also export a file with all the data after stopping a run using the Export CSV button.
   - Start an initial test collection for 20-30 minutes and see if the data looks good. Remember, you will take an overnight data set. This is your opportunity to ensure that you have set your parameters accordingly.

Checkpoint **Initial Test Run:**↑ **Show you 20-30 minute run to a GSI. Do you trigger and output counts make sense? Does your distribution make sense? Adjust your initial parameters as needed.**

Using the settings you have demonstrated in the previous checkoff (first trigger level, second trigger level, etc), set up and execute an overnight or a weekend-long data collection run. A very large number of detection events will be needed in order to perform proper data analysis (100k to 1M pulses). Do your best to ensure the muon decays are being counted correctly before you leave for the night, as there is not enough time allotted to perform multiple tests.

7.5 Calibration of Electronics

In experimental physics it is important to understand the electronics you work with and the limitations they place on your data. The equipment can reveal a lot about the scope of the experiment. To better understand the setup for this lab you will need to run three types of calibrations:

1. **Time resolution of the ADS**. Fundamentally, the ADS is taking an analog signal and turning it into a digital one, i.e., it is acting as a *digitizer*. Our digitizer works in discrete steps and therefore there is a limit on its time resolution that is based on how quickly it can process and convert the incoming signal. We will want to determine this time resolution. It is important to stop and think why we care so much about the time resolution of the digitizer. What timescale should it be on to ensure that our experimental procedure works and the data we receive is accurate? (Hint: What are we measuring?) The time recorded by the two PMTs for *through-going muons*, (i.e. muons that go through both halves of the scintillator tank) will give us an understanding of the timescales that the experiment operates on. In order to extract this information from the **Muon Detection Program** the correct number of

expected pulses in each channel should be input. What should these values be (Hint: The first PMT is located above the scintillating material while the other is located below)? Use the **View Signals** program to make sure that the trigger level and pulse thresholds counting rate is reasonable; the trigger level should be roughly near 0.1V (you may need to apply additional cuts on pulse amplitude if you set the thresholds too low). After you have collected a sufficient amount of data you are ready to examine the response rate between the two channels. This time difference data has some distribution. Fit it to a function you think is suitable (Hint: Here you have a large number of data points whose values are independent to one another all with small variations about an expected value. There is a common theorem used in statistical distributions such as this and will help dictate which function will fit the distribution. What two things can contribute to this?) You should see that the mean is not centered on zero, why is this? What does the standard deviation of the function tell you? (Hint: The standard deviation corresponds to the margin of error within your distribution. What are we measuring and what does the error tell us?) Do you need to apply any correction to the lifetime data? **Note: as of August 2026, the bottom PMT is not operational. We are hoping to fix this soon. Check with a staff member whether you should do anything for this step of the experiment. All steps below are still perfectly functional and should be performed.**

Checkpoint **Time Difference Data:**[↑] **Which function did you fit to the time difference data? Explain what the mean and standard deviation tell you about the data and its physical significance.**

2. **Efficiency of the digitizer**. Since you will determine the muon lifetime from a histogram of *counts vs time*, you will need to make sure that each time interval is counted by the apparatus equally. To check this, you need to present the apparatus with *a-priori* uniform distribution of time differences. This is done by combining signals from two *uncorrelated* sources: a through-going muon, and an external pulse generator (DG645). As long as the counting rates from the two sources are similar, it's likely that the two trigger pulses will come from different sources.

   - First, set up the apparatus according to the procedure: **Calibration setup Procedure for Muon Lifetime**.
   - Now use a ZSCJ-2-1 Power Splitter to add the Delay Generator signal to the Muon pulse, and run it to Channel 0 of the ADS. The Power Splitter takes a signal from input 1, inverts it, and adds it to input 2. It outputs the sum on S. Specifically: plug the AB output of the Delay Generator into input 1 of the power splitter, plug the amplified muon signal into input 2 (it does not matter if you use the top or bottom chamber), and then plug Channel 0 of the ADS into output S of the Power Splitter.
   - You can use the oscilloscope to check the signal. Do you see the delay generator peaks? The muon signal peaks? The polarity of both should be the same (be sure to remember if the polarity is positive or negative). If you wish to change the polarity of the delay generator pulses, press SHIFT and then press either 8 or 9. 8 corresponds to positive polarity, which 9 to negative.
   - We would like to trigger on the muon events and record when the pulses of the pulse generator arrive in relation to that. For this, we use the Muon Detection Program's **Live Histogram** tab. Remember to select the appropriate attenuation based on the signal polarity! Set the first trigger level to be above the amplitude of the Delay Generator peaks to avoid triggering on it, but use a second trigger level less than that. For this, check on the oscilloscope for amplitudes of typical muon and pulse generate events. Since the muon events may be of low amplitude you may need to reduce the output of the pulse generate with an attenuator. Set the number of pulses to be 2 on Channel 0 and 0 on Channel 1 (or keep channel 1 off).

     Note: you may also try the opposite way, i.e. triggering on the pulse generator and looking at the muons in relation to that.
   - Check the data rate: what do you expect?
   - Collect the data: a total of more than 100k output pulses would be more than enough.
   - Save your data, and repeat the procedure for Channel 1 of the digitizer. **As of August 2026, the bottom PMT is not working. Instead, you can add two pulses from the delay**

generator in order to simulate both muon pulses as well. Check with a staff member what your procedure should be for this step.

- Data analysis: look at the distribution of time differences dt in each channel. Apply selection on pulse heights if the thresholds are set too low (e.g. the second pulse in each channel should be from DG645, and therefore very constant in amplitude). Select the interval where the counting rate is most uniform. Determine if the data are consistent with constant counting rate, or if you need to make a correction to the real muon decay data.
  (a) Plot the histogram of counting rate vs dt for each channel. Remove bins that obviously over-count, if any.
  (b) If there is an obvious slope to the distribution, you may not have set the thresholds correctly, and the VI is triggering on two muon pulses, instead of one muon and one generated pulse. Try adjusting thresholds in the Python program after carefully inspecting the amplitudes of the muon and generator pulses. The generator pulse has a very stable amplitude, typically between 0.3-0.4 V after the digital filter.
  (c) Once you have a reasonable distribution of counting rate vs dt, convert it to semi-log scale, and fit to a straight line. Is the slope significant? If it is, how does it affect your muon data?

3. **Calibration of the digitizer clock**. Even though the Python software reports the time interval dt in seconds, you need to check that the time scale is accurate. To do this, you need to send pairs of pulses from DG645 generator to the digitizer, vary the time interval between the pulses, and reconstruct this time interval with the Python software.
   - First, reconfigure the internal trigger of the Delay Generator to 50 Hz.
   - Set an AB pulse of 1V and CD pulse of 0.5V. Configure the CD output of the Delay Generator to send out a 100ns pulse some time after the AB pulse. Remember that you set the AB pulse to be sent 1 μs after the internal trigger, so setting the CD pulse to begin at 3 μs will correspond to a time difference of 2 μs between the pulses. Set the AB polarity to negative and the CD to positive (remember the splitter inverts the Channel 1 input).
   - View the signal on the scope and make sure that the signal is what you expect. Note the time difference between the pulses. Use the scope to determine the initial trigger (you should trigger on the AB pulse) and the pulse detection threshold (CD pulse).
   - Plug output AB into input 1 of the power splitter, and plug output CD into input 2 of the power splitter. Plug output S into the digitizer channel to be tested
   - Run the **Live Histogram** tab of the Muon Detection Program program to collect data. Set your trigger and pulse detection thresholds to what you determined when you viewed the signals on the scope.
   - You should see just one of the histograms bins being filled. You should collect data for enough time differences to be sure that the digitizer reads all time differences correctly, from 1 μs to 40 μs.
   - Repeat this procedure for the other digitizer channel.

4. **Linearity of the pulse height measurement**. With the setup above, set the time difference between **AB** and **CD** to some fixed value (e.g. 5 μs), and vary the amplitude of **AB** and **CD**. Collect data for each setting, plot the reconstructed pulse heights versus the amplitude set by the DG645 generator. How linear is the digitizer? Can you also think of a way to check the linearity of the 40 dB amplifier?

7.6 Additional investigations

Cosmic ray physics is a rich subject, and we'd like to give you the opportunity to explore some of its other facets here. **Pick one of the investigations below, develop a plan for investigating it, and perform it.** There is a lot of room here for creativity! The Python program allows you to explore a variety of other

pieces of data that we haven't used so far; see what is available and refer to some of the references above for ideas as well.

Optimal filtering

One option the Python program provides is to save the raw traces, along with the pulse characteristics. Doing this rapidly accumulates extremely large amounts of data, so normally the traces are not saved. However, access to the raw traces opens up a number of interesting analysis possibilities. Perform a *short* run (begin with a few minutes to get a sense of the quantities of data) with the save traces checkbox selected. Then, investigate constructing an optimal filter. To do this, you should examine the noise power spectrum of the detector, apply it to your pulses, and compare the signal to noise ratios before and after your filter. You should provide a brief discussion of the theory behind this type of filtering, which is broadly used in particle physics. See references above.

Atmospheric and space weather effects on muon flux

The atmosphere has an effect on muon count rate, in particular temperature and air pressure. These are also, naturally, dependent on time of day. Search these effects, and attempt to calibrate for them. A useful Python package is meteostat, which allows you to access live data from local weather stations. Any data you collect this way should be shared with staff - we are attempting to build up a longer term dataset so that you and future students can look for trends more easily. How can you check whether there are statistically significant correlations between count rate and atmospheric variables? (Begin with looking at the concept of correlograms.) Just as with atmospheric effects, muon flux is also affected by the solar cycle and space weather, which NASA provides live data on (you can also look at Python packages like solarpy or sunpy). (Of course, to really see these effects, one must also calibrate for atmospheric effects.)

Energy distribution of muons

We've focused on the lifetime of muons and the muon count rate, but we can also examine the energy distribution of muons. What energy distribution of muons do we expect at our altitude? How can we tell what energy distribution we see using the data collected your muon lifetime investigations? What are confounding factors?

Beyond muons

We've been focusing on muons by looking for the decay pulse, but what about traces for which there is a first pulse but no decay pulse? Perform one of the above investigations but considering overall charged particle detection (this can also help you get a larger dataset). You should discuss what types of particles you might be seeing, how you know, and provide some discussion of cosmic rays more generally.

Other

Investigation of your choosing. Confirm your investigation topic with a professor before proceeding.

Checkpoint **Additional investigation.:**[↑] **Explain the topic of your additional investigation to a GSI/professor and your plan for the measurements you will take.**

8 Analysis

Excel is insufficient for this kind of data analysis. You should write a script in Python, MatLab, ROOT, etc.

1. Import the raw data from the csv into a program of your choice. Open the text file to see column delineations. Pay attention to column heads. A good place to start is by making an array/list for each column. (In Python, you may wish to use the 'pandas' package.)

2. Inspect the distribution of pulse heights and pulse widths for both the first and second pulses. Can you explain their features? Apply a cut, which means to remove outliers (such as low-amplitude noise pulses). See how the distribution of time separation dt changes when you cut on pulse height.

3. Determine if the dt plot needs to corrected for any non-uniformity measured in **Step 2** of the **Cali-**

bration section

4. Determine if you should apply time calibration from **Step 3** of the **Calibration** section

5. Make a semi-log plot of counts D vs. time separation dt. On this plot, fit the background (which starts from about 10 to 20 μs) to a straight line (with the slope not necessarily zero). Extrapolate the line under the muon decay region and subtract the background in each channel from D, yielding D_2. **THINK:** How does the choice of the background fit influence your results? Is the fit to the background a constant throughout time, a linear decay, or an exponential decay. Once you have determined the fit and keeping in mind that we are interested in the counting rate, think about how you can properly remove this background noise from the muon decay region. Discuss what could be determining the fit of the background and its correlation to the data you are interested in.

6. Make a semi-log plot of D_2 vs dt. Make sure errors are correctly computed for $\log D_2$. Make a least-squares fit to a straight line (over a time interval where such a fit makes sense). The slope of this line is inversely proportional to the "effective" muon lifetime. Also show lines corresponding to "effective" lifetimes that are one standard deviation higher and lower than the mean lifetime. All lines should be normalized to the same total number of counts. After you make a fit, also check and see how the calculated exponential curve fits the measured curve. Observe how the lifetime values and the χ^2 of the fit change depending on fit range. Justify why you picked a certain fit range.

7. Be sure to read Evans Chapter 15 at this point. As indicated in the lab write-up, some negatively charged muons (μ^-) undergo nuclear capture in the muonic carbon nuclei of the mineral oil in the scintillation tank. These muons generate a start pulse when they enter the detector but do not generate a stop pulse when they are nuclear captured. Therefore their decay times are not detected. Although the reprints show that the entire population of $\lambda_{\text{combined}} = \lambda_{\text{decay}} + \lambda_{\text{captured}}$, it may not be obvious that the *free-decaying* μ^-'s will also decay with this combined rate. This is a subtle point that you should be sure to ask about if you need to.

   - Note that $\lambda_{\text{decay}} \cong \frac{1}{2.2\mu\text{s}}$ for both positive and negative free muons, and that $\lambda_{\text{captured}} \cong \frac{1}{26\mu\text{s}}$ is the rate of negative muon decay in the material (carbon) of the detector.
   - As discussed above, there are *two* different populations of muons that you measure, μ^- and μ^+, and they decay with the different decay rates, $\lambda^{\mu^-}_{\text{combined}}$ and $\lambda^{\mu^+}_{\text{decay}}$ respectively. The histogram of decay-times that you will get from the equipment (barring background and noise) is the sum of *two* exponentials: $\text{Histogram}(t) = A\exp(-t\lambda^{\mu^-}_{\text{combined}}) + B\exp(-t\lambda^{\mu^+}_{\text{decay}})$, where A and B are constants. To analyze these data properly requires a non-linear curve fitting program. But because $\lambda^{\mu^-}_{\text{combined}}$ and $\lambda^{\mu^+}_{\text{decay}}$ are very close to each other, it is difficult to get good results. Instead, it is recommended that you fit your data to a *single* exponential: $\text{Histogram}(t) = Ce^{-t\lambda_{\text{effective}}}$. This gives you an "effective" life-time. The use of the word "effective" is only to signify the answer the computer gave you. It is possible to fit any curve to a single exponential and get an answer; the computer just does the best it can. So when you are asked to estimate the true lifetime, expect to make some approximations and don't expect to come up with an exact answer.
   - Assuming that half of the observed decays come from μ^-, with apparent decay rate $\lambda^{\mu^-}_{\text{combined}}$, estimate the magnitude and sign of the difference between the true muon lifetime and the "effective" lifetime (for μ^+ and μ^- together) measured in this experiment.

8. What is your value of the muon lifetime and its uncertainty, and how does this value compare with the published lifetime? **IMPORTANT:** estimate *systematic* uncertainties, i.e. think how the procedure, calibration, measurement apparatus and resolution settings, etc. may bias your results.

9. What is the value and uncertainty of the Fermi Coupling constant determined in this experiment? How does this measurement compare with the published value?

8.1 Note on Analysis Tools

This analysis can be done with a variety of tools, but we would recommend one of the more common packages designed for high-volume datasets. Thus, while it's possible to do a rudimentary version of the analysis in Excel, we do not recommend it, and we do not provide pre-packaged histograms of dt for that reason. Python is installed on the advanced lab computers, including the helpful packages MatPlotLib, NumPy, SciPy, and Pandas, and there is extensive documentation available online. Comprehensive tutorials for MATLAB are available on the web (some useful functions are `hist()` and `histfit()`). You have access to MatLab and Mathematica for free as a Berkeley student. In any case, it would be wise to review EAX.

The tool of choice for High Energy and Nuclear physics is **ROOT**. It is a sophisticated data statistical analysis package, specifically designed for analyzing large data sets, curve fitting, etc. ROOT is most natural for reading a large data set, applying cuts, and performing non-linear fits. ROOT is available for free download from **CERN**, and there is good documentation and tutorials on the **Root's main site**. ROOT can also be used with Python (and installed via conda as outlined in the link above - unfortunately as of 2026 it is not installable via pip unless you are on Linux and willing to use an experimental wheel) - see this as a starting point. We also provide some sample (bare bones) analysis examples below in C++: **Muon Analysis in ROOT**.

9 Computer simulation of experiment and analysis

Write a program in Python (or other software of your choice) to simulate muon decays and accumulate their lifetime distribution. Fit the generated data using whatever means you used above to determine the muon lifetime. Software routines that simulate data are called Monte Carlo programs (see ref. 10 or 11). By fitting this computer-generated data you can directly verify that the fitting procedure reproduces the input parameters (i. e., number of events and lifetime). These Monte Carlo simulations are common in complex experiments to model the experimental system and check the analysis methods.

Computers usually generate random numbers r_i uniformly distributed over the interval (0,1), that is, the successive values, called iterates, of r_i are described by the probability distribution

$$P(x)dx = 1 \tag{4}$$

You wish to simulate a series of muons whose lifetimes t_i follow an exponential probability distribution

$$P(x)dx = \frac{1}{\tau}\exp(-\tau x) \tag{5}$$

where τ = mean muon lifetime. Prove that

$$t_i = -\tau \ln r_i \tag{6}$$

will convert iterates from the uniform distribution to the exponential distribution [see ref. 10 or 11]

Write a program to

- generate a number of exponential iterates, t_i, and
- sort them into time channels N_k, which roughly correspond to the width and range of the experiment channels. (Increment N_k by 1, if t_i falls into the range of channel k).
- and store the data on the disk drive.

Fit these data with your fitting program. Investigate the dependency of the fitting program output on the number of events simulated. How many exponential iterates are needed to get good results from the fit?

A more detailed Monte Carlo analysis of your experiment is possible. For instance, generate the accidental background and study the influence of accidentals, statistics and fit -range on the results of your analysis.

- Last day of the experiment please fill out the **Experiment Evaluation**

10 References

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5. C.S. Sutton et al., "**Undergraduate Cosmic Ray Muon Decay Experiments with Computer Interfacing**", Computers in Physics, Nov/Dec, 76 (1987), #QC52.C658.
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8. National Instruments **Waveform 5124 Digitizer Manual.**
9. **RCA Photo Multiplier Tube Manual**.
10. **Stanford Research Systems DG645 Manual**
11. Full equipment list found **here**